AEROCAST NCR

Air quality and weather report: Delhi

PM2.5 over the past 48 hours, the regional atmosphere, the data behind it and how far to trust it.

Place	Delhi (28.61 N, 77.21 E)
Period	Fri 09 Oct 04:30 IST to Sun 11 Oct 03:30 IST
Pollutant	PM2.5
Generated	Sun 11 Oct 04:28 IST
Report id	55aebd44d814

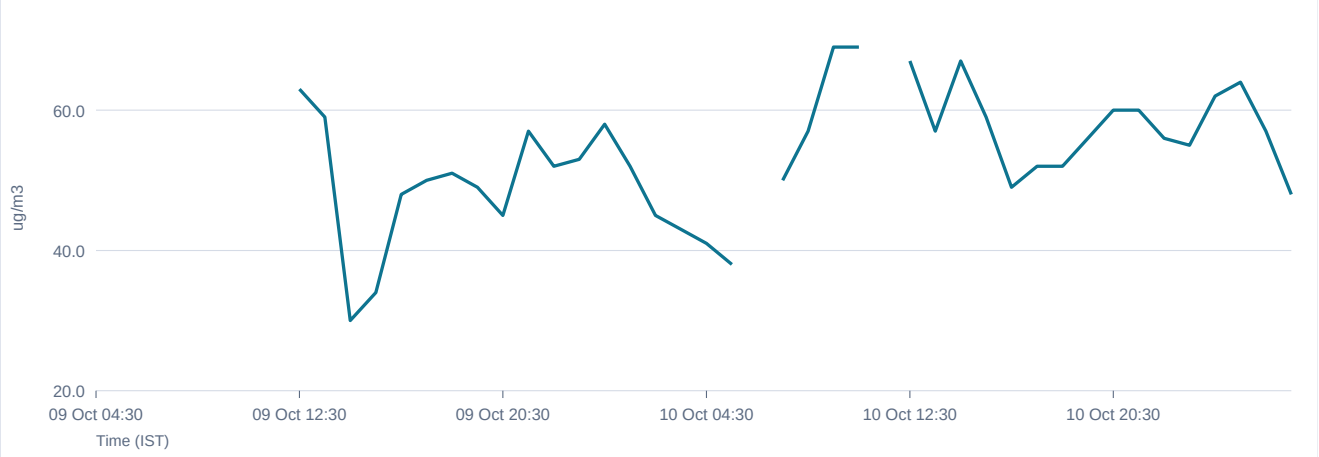
Summary

Every figure is labelled with what kind of number it is. An observation is a station measurement; a weather model or atmospheric model value is model output for a grid cell; a forecast is a prediction; a simulation is computed from assumptions. Gaps are shown as gaps and are never filled in.

- PM2.5 was observed at New Delhi (2.9 km from the Delhi centre) in 38 of 48 hours. The latest value was 48.0 ug/m3 at Sun 11 Oct 03:30 IST; the range was 30.0 to 69.0 ug/m3 with a mean of 53.5 ug/m3.
- The largest rise within 6 hours was +31.0 ug/m3 (82%), from 38.0 at Sat 10 Oct 05:30 IST to 69.0 at Sat 10 Oct 09:30 IST.
- Comparing the start and end of the period, the mean moved from 48.0 to 57.6 ug/m3 (+9.6 ug/m3).
- Weather-model values for the period: temperature 23.5 to 34.0 °C; wind speed averaging 9.2 km/h; boundary layer height 180 to 2410 m; 1.2 mm of rain in 5 hours.
- During the largest rise, wind speed averaged 10.0 km/h (above the period mean of 9.2); boundary layer height averaged 442.0 m (below the period mean of 657.1); temperature averaged 25.2 °C (below the period mean of 27.4); relative humidity averaged 83.8 % (above the period mean of 70.6). These are conditions that coincided in time with the rise. They do not show what caused it.

PM2.5 observations at Delhi

Station New Delhi, 2.9 km from the centre of Delhi (provider: AirNow). Observation, source [A1].

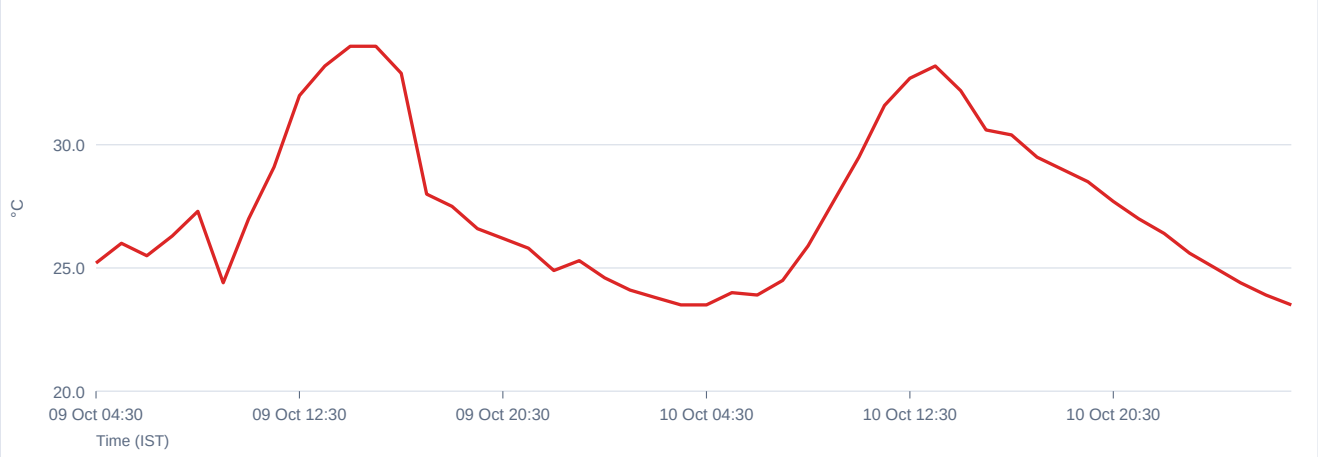


Hourly PM2.5 at New Delhi, observation, IST. A missing hour is a gap in the line.

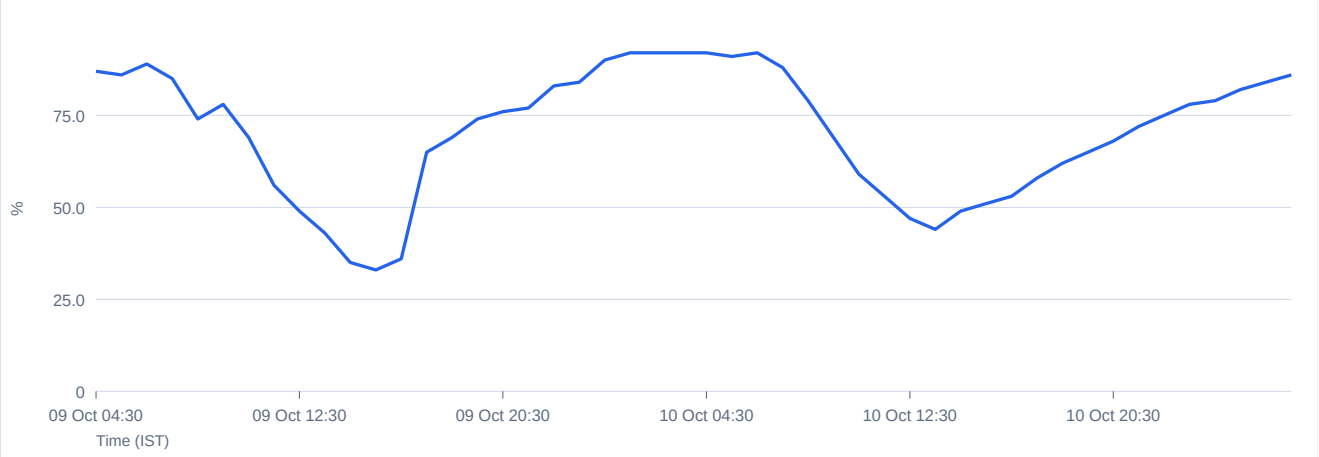
Measure	Value
Hours observed	38 of 48 (10 missing)
Latest	48.0 ug/m3 at Sun 11 Oct 03:30 IST
Lowest	30.0 ug/m3 at Fri 09 Oct 14:30 IST
Highest	69.0 ug/m3 at Sat 10 Oct 09:30 IST
Mean	53.5 ug/m3
Largest 6-hour rise	+31.0 ug/m3: 38.0 at Sat 10 Oct 05:30 IST to 69.0 at Sat 10 Oct 09:30 IST
Trend	48.0 to 57.6 ug/m3 (+9.6); mean of the first 16 h versus the last 16 h of the period (observed hours only)

Weather over the period

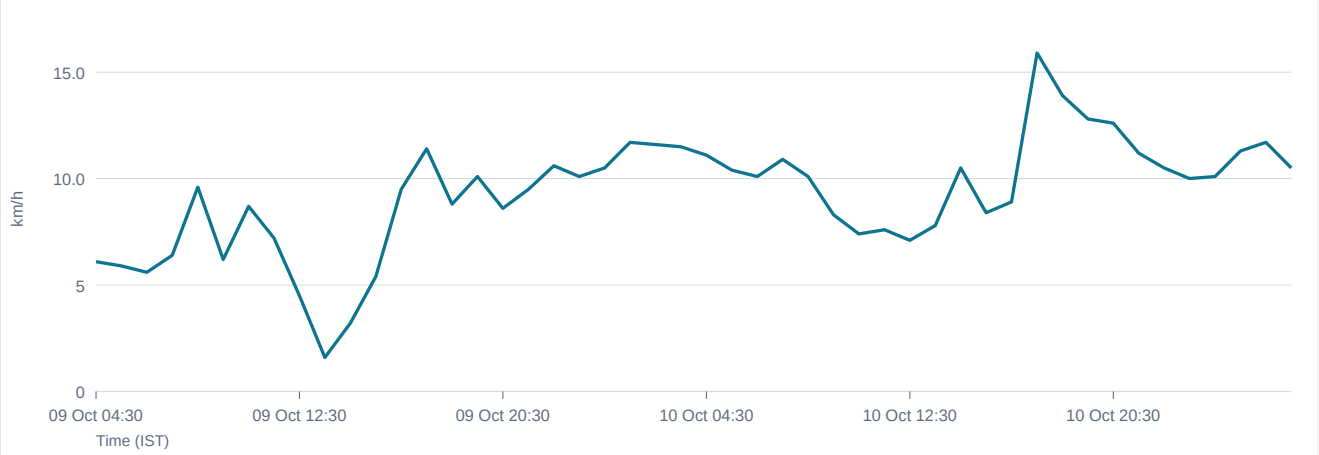
Weather-model values for the grid cell at 28.58 N, 77.19 E (elevation 217.0 m). Weather model, not station measurements. Source [W1].



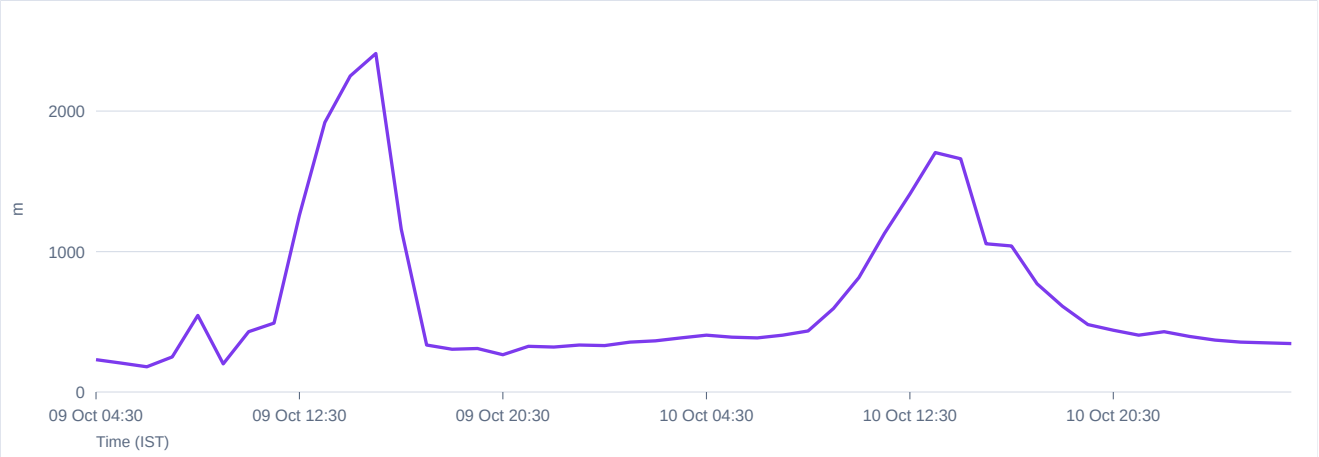
Temperature, weather model, hourly, IST.



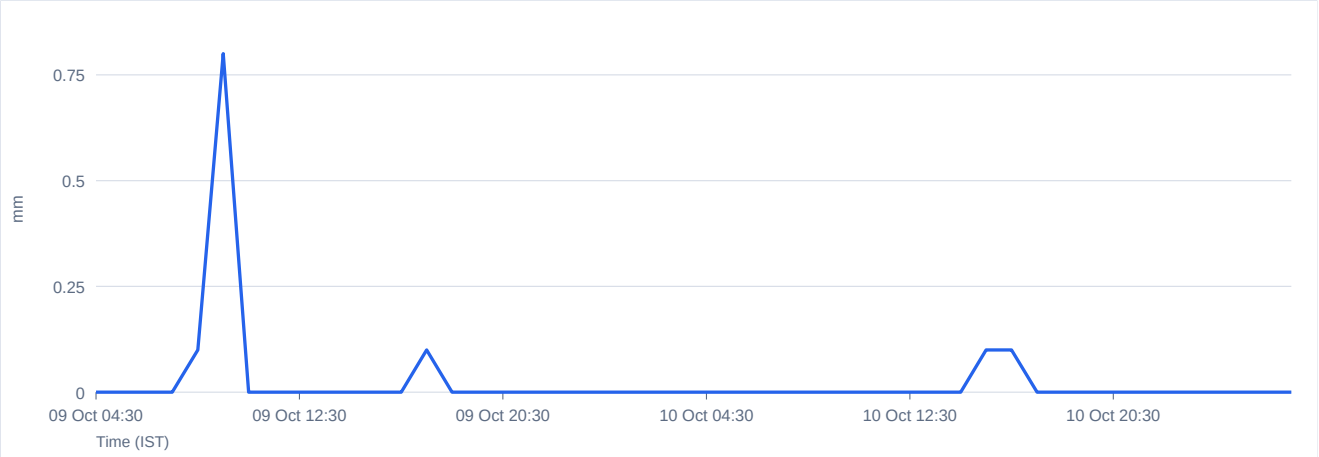
Relative humidity, weather model, hourly, IST.



Wind speed at 10 m, weather model, hourly, IST.



Boundary-layer height, weather model, hourly, IST.



Hourly precipitation, weather model. Total 1.2 mm in 5 rainy hours.

Variable	Min	Max	Mean	Unit
Temperature	23.5	34.0	27.4	°C
Relative humidity	33.0	92.0	70.6	%
Dew point	15.5	23.6	21.0	°C
Wind speed	1.6	15.9	9.2	km/h
Surface pressure	985.3	990.4	987.4	hPa
Cloud cover	0.0	100.0	33.4	%
Boundary layer height	180.0	2410.0	657.1	m
Precipitation	-	-	1.2 total	mm, 5 rainy hours

Atmospheric conditions and what coincided with the largest rise

- Ventilation index (boundary-layer height times wind speed): latest 1006 m2/s, mean 1490 m2/s.
- Hours since rain: 11.

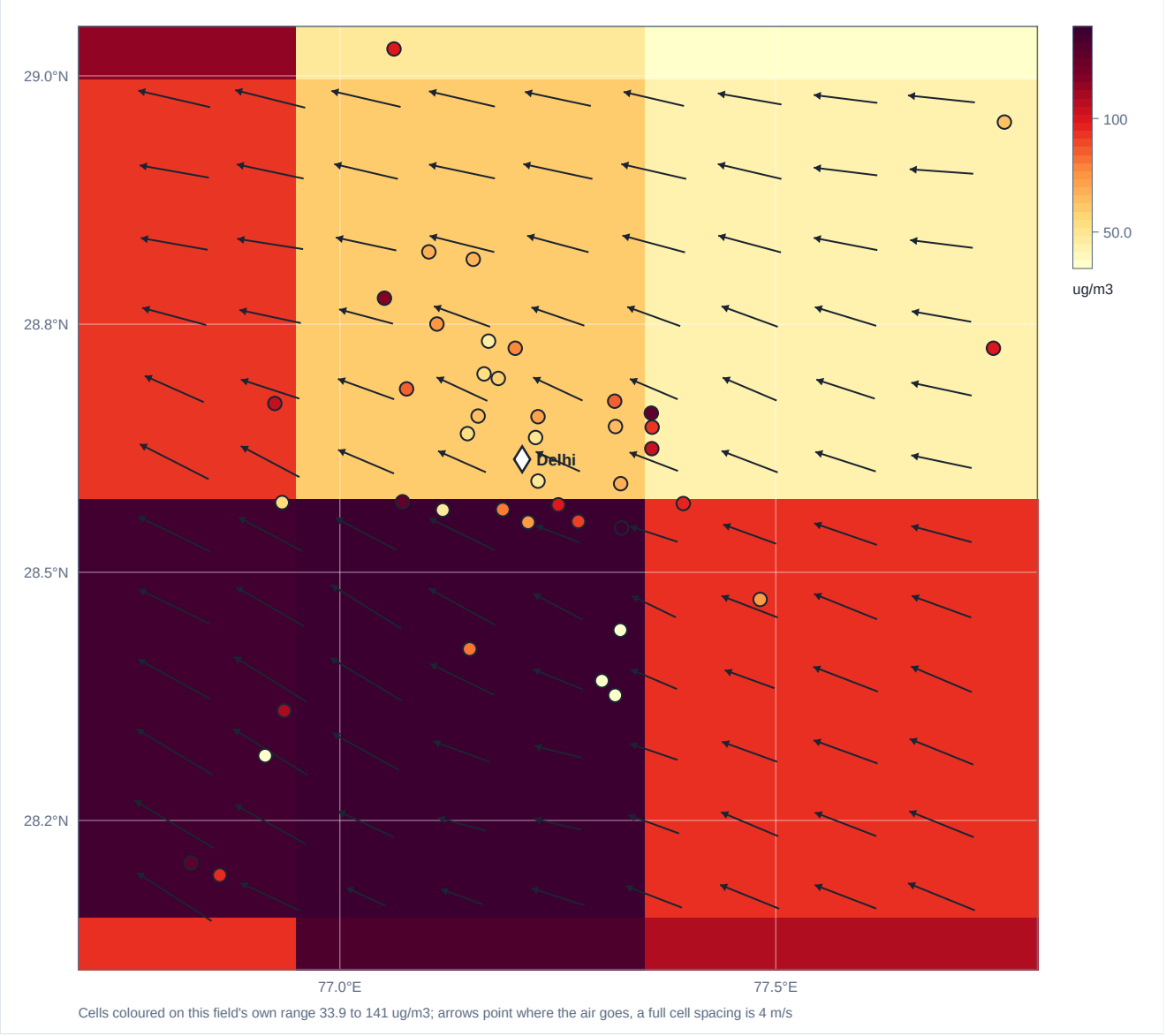
Descriptive indicators computed by simple arithmetic from weather-model values. They are not measurements, carry no thresholds or risk labels, and do not show whether a temperature inversion exists.

Weather during the largest rise (Sat 10 Oct 05:30 IST to Sat 10 Oct 09:30 IST)

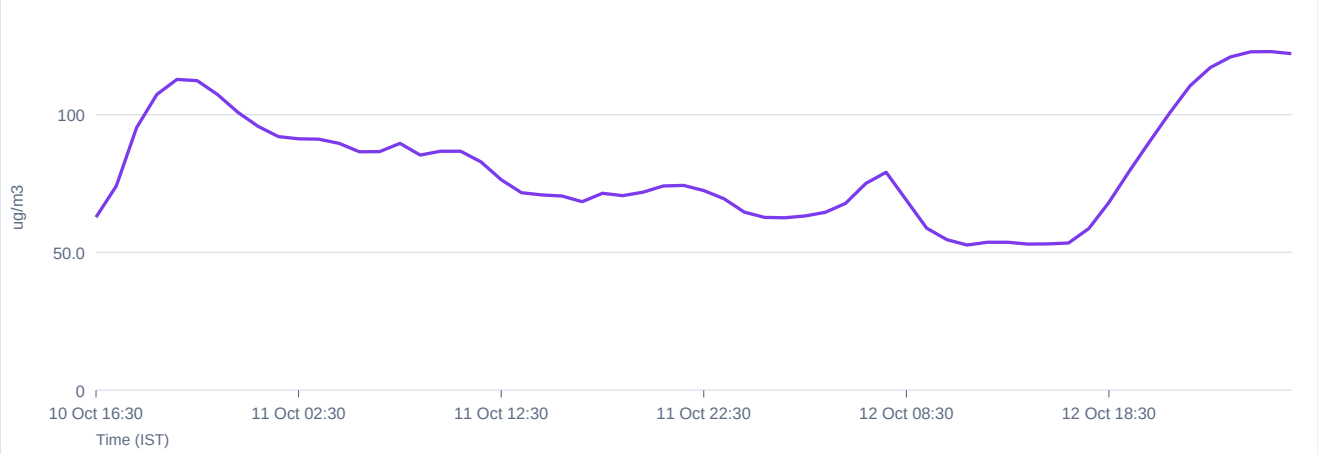
Variable	During the rise	Period mean	Relation
Wind speed	10.0 km/h	9.2 km/h	above
Boundary layer height	442.0 m	657.1 m	below
Temperature	25.2 °C	27.4 °C	below
Relative humidity	83.8 %	70.6 %	above
Precipitation (total)	0.0 mm	-	

These are conditions that coincided in time with the rise. They do not show what caused it.

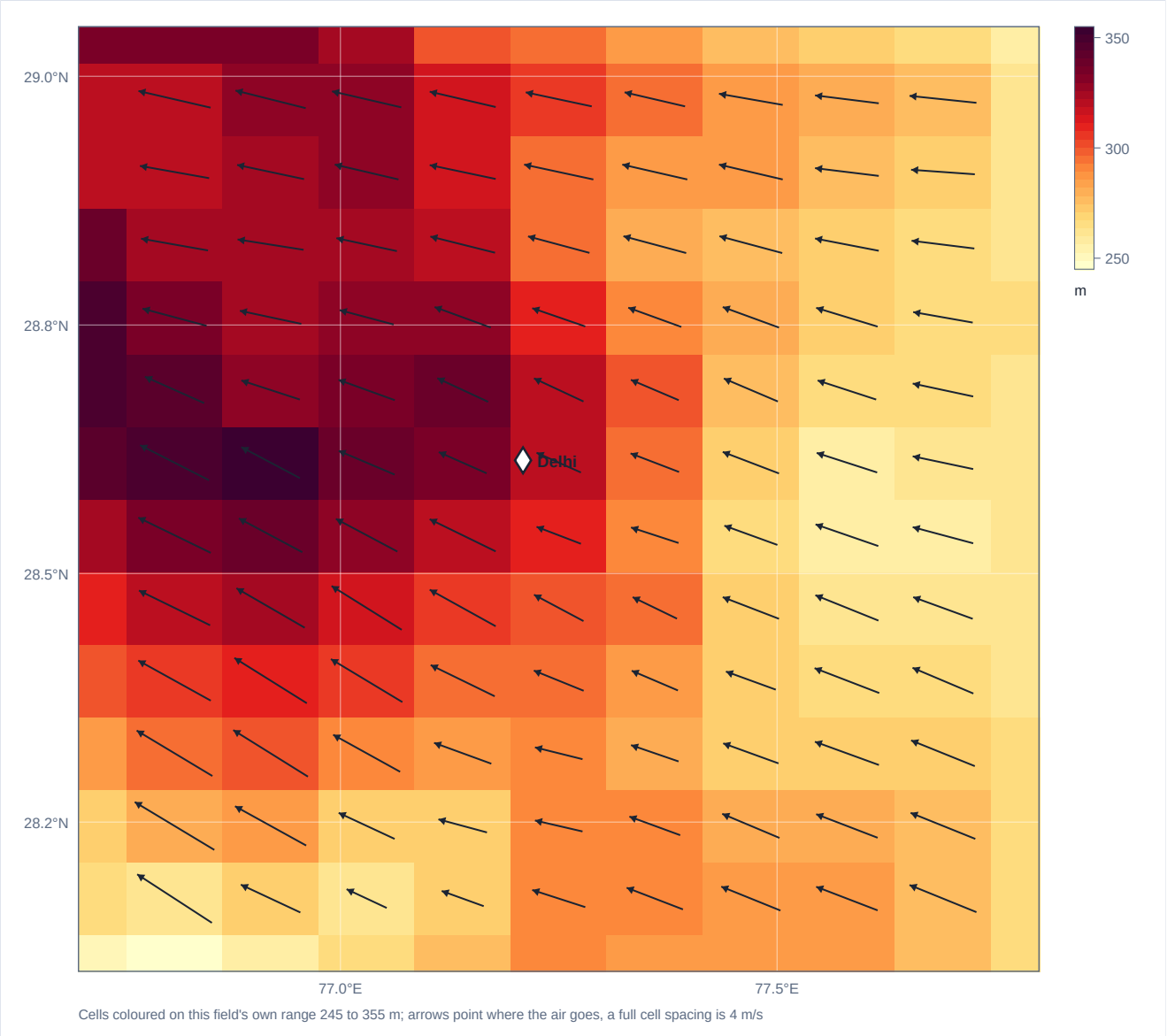
The regional picture: air and wind across Delhi-NCR



PM2.5 on the atmospheric-model lattice at Sun 11 Oct 04:30 IST (atmospheric model, not a measurement). Cells are about 0.4 degrees (44 km), so the field is a few large blocks. Dots are the newest station readings on the same colour scale (observations, of varying age); arrows are the weather model's 10 m wind.

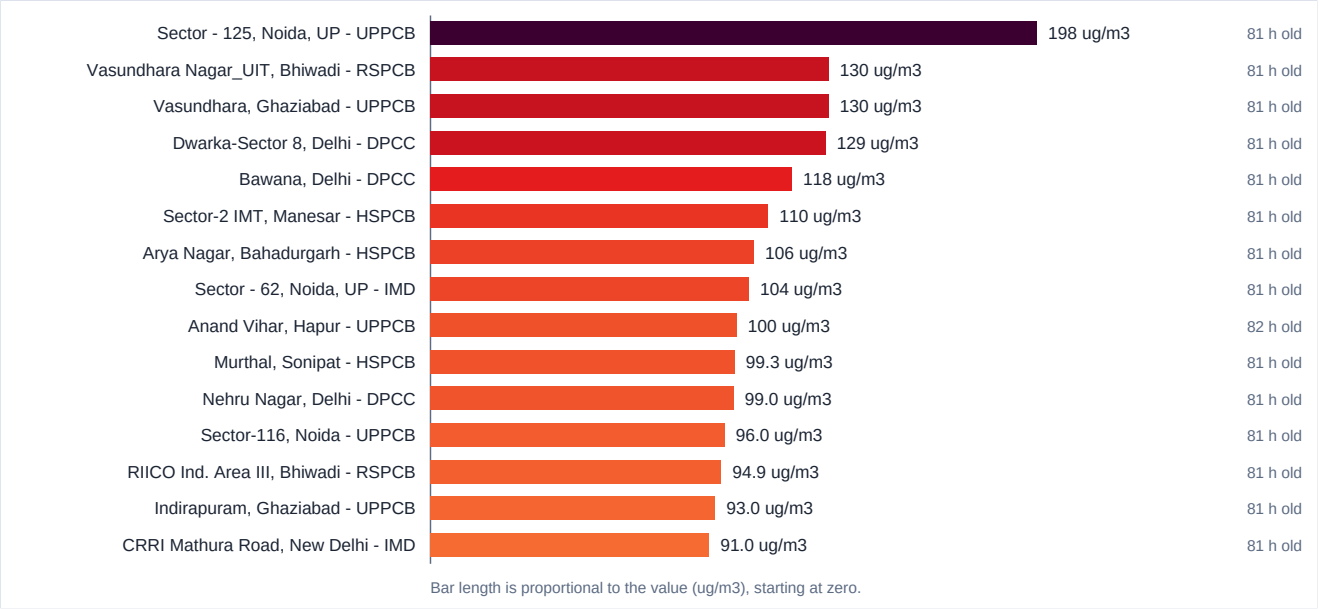


Mean of the atmospheric-model lattice over Delhi-NCR, 12 hours back to 48 hours ahead (atmospheric model, forecast beyond now), IST.



Boundary-layer height, weather model, same hour. A shallow layer holds emissions in a small volume; a deep one dilutes them.

The station network: latest PM2.5



The 15 highest of 42 stations reporting PM2.5: newest observed hour at each, in the sensor's unit (observations). The median reading is 81 hours old, so these are not 'now'.

HOW CURRENT IS THIS?

The station feed lags: the median newest reading above is 81 hours old and the oldest is 127 hours old. Compare with the atmospheric model above, which is current but coarse.

Forecast

NOT AVAILABLE

Not enough recent data to forecast: the newest observation near the location is 26 h old (the limit is 6 h).

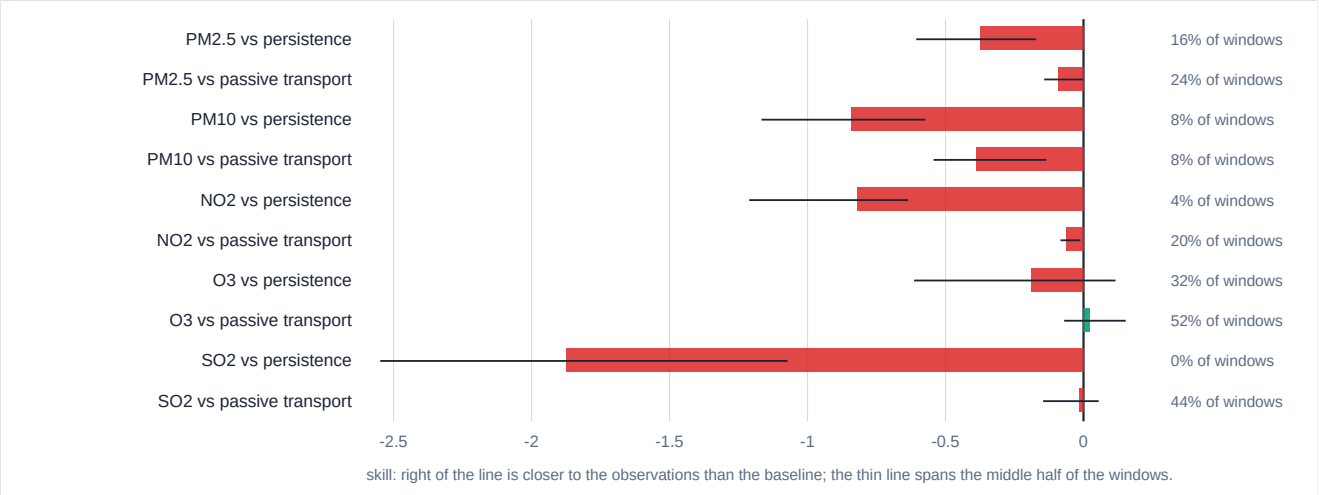
How well the models perform

Trained model run 4702878405a8, created 2026-10-10. Errors are mean absolute error in ug/m3 on a held-out later period.

Pollutant	Horizon	Model	Model error	Persistence error	Beats persistence
NO2	+1 h	lightgbm+random_forest	7.5	7.9	yes
NO2	+6 h	random_forest	12.9	18.5	yes
NO2	+24 h	random_forest	13.9	15.2	yes
O3	+1 h	random_forest	3.9	4.7	yes
O3	+6 h	random_forest	7.7	14.3	yes
O3	+24 h	random_forest	7.7	8.1	yes
PM10	+1 h	random_forest	16.7	17.0	yes
PM10	+6 h	random_forest+lightgbm+xgboost+svr	34.3	39.8	yes
PM10	+24 h	random_forest	42.4	41.4	no
PM2.5	+1 h	random_forest	8.5	8.7	yes
PM2.5	+6 h	lightgbm+random_forest	15.1	17.9	yes
PM2.5	+24 h	random_forest	17.5	18.5	yes

Persistence repeats the last observed value. A model is only useful where it beats it.

The coupled transport model against simple baselines



Hindcast of the coupled model (defaults (deposition + washout + oxidation)) over 25 historical 12-hour windows, scored at stations withheld from the starting field (simulation evaluated against observations).

Skill is MAE at stations withheld from the initial field, leads 1..N h; skill = 1 - MAE_model / MAE_baseline (positive: closer to the observations). Negative values mean the model did worse than the baseline. The coupled model has no emissions and did not beat persistence; it is a picture of how air moves, not a forecaster.

What data the application has, and what it does not

8 available, 5 stale, 6 obtainable, 9 missing items are tracked. A gap is shown as a gap. Where data are missing the application says so and does not substitute invented or synthetic values.

Air-quality observations

Data	Status	What exists	What a gap prevents
PM2.5 station observations	stale	2021-12-31 to 2026-10-09; newest well-covered hour 2026-10-07T14:00Z, 81 h ago; about 68 stations.	The newest stored hour is old, so every map built from observations is a replay, and live forecasts are refused (inputs older than the 6 h limit).
PM10 station observations	stale	2021-12-31 to 2026-10-07; newest well-covered hour 2026-10-07T14:00Z, 81 h ago; about 68 stations.	The newest stored hour is old, so every map built from observations is a replay, and live forecasts are refused (inputs older than the 6 h limit).
NO2 station observations	stale	2021-12-31 to 2026-10-07; newest well-covered hour 2026-10-07T14:00Z, 81 h ago; about 68 stations.	The newest stored hour is old, so every map built from observations is a replay, and live forecasts are refused (inputs older than the 6 h limit).
O3 station observations	stale	2021-12-31 to 2026-10-07; newest well-covered hour 2026-10-07T14:00Z, 81 h ago; about 68 stations.	The newest stored hour is old, so every map built from observations is a replay, and live forecasts are refused (inputs older than the 6 h limit).
SO2 station observations	stale	2021-12-31 to 2026-10-07; newest well-covered hour 2026-10-07T14:00Z, 81 h ago; about 68 stations.	The newest stored hour is old, so every map built from observations is a replay, and live forecasts are refused (inputs older than the 6 h limit).
Live OpenAQ feed	available	Reachable, key accepted	Many stations publish with a lag of days, so 'live' is rarely the last hour.
CO observations	missing	All CO sensors in the window are labelled ppb with a median near 1: physically impossible, so they are excluded (UNIT_SUSPECT).	No CO in any map, model or coupled run.

Data	Status	What exists	What a gap prevents
NO and NOx observations	obtainable	Offered by the OpenAQ feed this application already uses (seen in the 2026-10-11 audit) but not ingested, so every model and map works without them.	O3 and NO2 cannot be coupled chemically; Ox = O3 + NO2 is used instead.
NH3 and VOC observations	missing	Not offered by any feed used here.	No nitrate or secondary organic aerosol in any model.
PM chemical speciation (sulfate, nitrate, OC, EC, dust)	missing	Only total PM2.5 and PM10 mass are observed.	The model cannot tell dust from smoke from sulfate; SO2 -> sulfate is a mass-conversion assumption.
Vertical profile of pollutants (lidar, sondes, tall-tower)	missing	All concentrations are at ground level.	The 3D view shows no measured concentration above ground; any layer drawn in 3D is labelled illustrative.
Satellite AOD and NO2 columns	obtainable	Not ingested. CAMS gives a modelled AOD only.	Air entering the domain is 'unknown'; no observed regional field fills the gaps between stations.

Meteorology and geography

Data	Status	What exists	What a gap prevents
Weather-model fields (temperature, humidity, pressure, rain, cloud, boundary-layer height, radiation, visibility)	available	Reachable. Native cells about 0.07 x 0.10 degrees (8 x 10 km), hourly, past days and 3 days ahead.	They are model values: sharper than the grid they come from they are not.
Winds aloft (80, 120, 180 m and 925, 850, 700 hPa)	available	Seven model levels from 10 m to about 3 km, with their geopotential heights.	Between those levels the 3D flow is interpolated, not resolved; there is no vertical wind (w) in the displayed flow.
Observed meteorology at the stations (temperature, humidity, wind speed and direction)	obtainable	OpenAQ lists temperature, relative humidity, wind speed and wind direction sensors on CPCB stations (2026-10-11 audit). They are not ingested: every weather value in the application is model output.	Errors of the model wind propagate into every transport result, unquantified.
Radar and live satellite cloud/rain imagery	missing	Rain and cloud are model forecasts; the Sentinel-2 background is a 2020 composite.	The picture cannot show what is in the sky right now.
High-resolution regional weather model (WRF, ICON-D2) for NCR	missing	Best available resolution here is about 8-10 km.	Wind features smaller than ~10 km, the Aravalli ridge's flow effects and city heat-island circulation are not resolved.
Terrain and satellite imagery	available	Cached on this machine.	Static: not today's imagery.

Sources and drivers

Data	Status	What exists	What a gap prevents
Emission inventory (traffic, industry, power, residential, construction)	obtainable	No source term exists anywhere in the application.	No model here can create pollution: forecasts are carried-air forecasts or statistical (ML) forecasts. This is the largest gap.
Fire and stubble-burning detections	obtainable	Not ingested.	The model cannot know about upwind smoke before it arrives at a station.
Traffic counts and congestion	missing	Not available.	Morning/evening peaks are not represented.
Land use, population density and building height	obtainable	Not loaded. The exposure tables exist in the schema and are empty.	Deposition velocity is one value per species; exposure cannot be population-weighted.

Models and knowledge

Data	Status	What exists	What a gap prevents
Trained pollutant forecast model	available	PM2.5, PM10, NO2, O3 at +1, +6, +24 h; trained 2026-10-10 (1 days ago).	SO2, CO and the +3/+12/+48/+72 h horizons have no model. PM10 +24 h does not beat persistence.
CAMS chemistry-transport model fields (via Open-Meteo)	available	PM2.5, PM10, NO2, SO2, O3, CO, NO, AOD, dust in cells of about 0.4 degrees (44 km: about 3 x 3 cells over NCR); model, not measurement.	Very coarse (about 44 km cells) and not assimilated with the stations here; it can differ strongly from them and shows no city-scale structure.
Coupled multi-species transport model	available	Five species with ventilation, deposition, wet scavenging and SO2 -> sulfate.	Not validated as a chemistry-transport model and has no emissions; the app shows its hindcast against persistence and passive transport.
Language model for the assistant	available	hermes3:8b installed	Without it the assistant and summaries report themselves unavailable.
Evidence documents (knowledge base)	missing	No documents are indexed.	Answers cannot cite guidelines or studies you have not added.

Health and exposure

Data	Status	What exists	What a gap prevents
Patient-level or hospital-admission data	missing	None provided; the exposure UI waits for it.	The application reports air quality, not health outcomes.

Data freshness, limitations and sources

Source	Status	Newest data (IST)	Detail
Weather-model values	fresh	Sun 11 Oct 04:15 IST	The provider's own data time.
PM2.5 observations	fresh	Sun 11 Oct 03:30 IST	Newest observed hour; readings older than 3 h are flagged stale.

What this report cannot say

- PM2.5 was observed in 38 of 48 hours; the other 10 hours have no data and are not filled in.
- Weather values are weather-model output for a grid cell, not measurements at the station.
- No scientific references are cited. The knowledge base has no searchable documents, so no references are cited.
- This report contains no forecast. Not enough recent data to forecast: the newest observation near the location is 26 h old (the limit is 6 h).
- It does not identify pollution sources: no emissions or source data were used.
- It cannot confirm a temperature inversion: vertical temperature data for this past period were not used.
- Concentrations are shown in the sensor's own unit and are not converted.

Sources

ID	Source	Retrieved (IST)
W1	Weather-model values, past 48 h. Open-Meteo, https://open-meteo.com/	Sun 11 Oct 04:28 IST
A1	PM2.5 at New Delhi. OpenAQ, https://openaq.org	Sun 11 Oct 04:28 IST

Background: the written reports

The reasoning, evidence and limits behind the figures above, as written in the application's Report page.

How the air of Delhi-NCR works, and how to read this application

> *Pollution in Delhi-NCR is not a single number. It is a field that the weather stirs, stacks up and sweeps away, and this application tries to show that field moving. This report explains the physical picture, what each page shows of it, and what it cannot yet see.*

Four ideas that explain most of what you see

Pollutants differ in where they come from and how they behave. Fine particles (PM2.5) are small enough to travel hundreds of kilometres and to reach deep into the lungs; they come from combustion, from dust that has been ground fine, and from gases that turn into particles in the air. Coarse particles (PM10 minus PM2.5) are mostly dust and settle out within a day or so. Nitrogen dioxide (NO2) is a combustion gas, strongest beside traffic and power plants. Ozone (O3) is not emitted at all: it is made in sunlight from other pollutants, and it is used up where fresh nitrogen oxides are plentiful, so it is often lowest in the most polluted places. Sulphur dioxide (SO2) comes from burning coal and oil and slowly becomes sulphate particles.

The weather decides how much of an emission you breathe. Emissions change slowly; concentrations change fast, because the volume of air they are mixed into changes. At night the air near the ground is a shallow, still layer and the same traffic and the same fires produce high concentrations. By mid-afternoon the sun has heated the ground, the mixing layer can be a kilometre or two deep, and the same emissions are spread through ten times the volume. This is why Delhi's PM2.5 almost always peaks in the early hours and falls to its daily minimum around 16:00 to 17:00, and why the boundary-layer height is one of the most useful numbers in the whole application.

Wind moves pollution, and the region matters as much as the city. Delhi sits at the south-western end of the Indo-Gangetic plain, a broad, flat basin that ventilates poorly. In October and November the winds turn north-westerly and smoke from crop burning in Punjab and Haryana arrives on them; in winter, calm and cold air under a temperature inversion lets everything accumulate for days; in the monsoon, rain washes particles out of the air and winds come in from the south-east. Dust from Rajasthan and the Thar arrives in the pre-monsoon months. A station reading is therefore the sum of what is local and what has been carried in.

Chemistry connects the pollutants. Sulphur dioxide and nitrogen oxides oxidise into sulphate and nitrate particles, so some of the PM2.5 you measure was a gas an hour or a day earlier. Ozone and nitrogen dioxide trade mass back and forth in sunlight. This is why the application can show a combined oxidant (Ox = O3 + NO2) that changes less than either does alone.

What each page shows

The Air Flow page is the main picture. Particles are carried along by the weather model's wind; each is coloured by the concentration of the air it is in, on the Central Pollution Control Board's index classes (green is good, dark red is severe). Underneath is a smooth concentration field. In 3D the same air moves at seven model heights from 10 m to about 3 km over real terrain. If you switch the source to Stored replay you see the last 72 hours of station observations; Live model shows the CAMS atmospheric model, which is the only source that is current; Coupled model carries a starting field forward with the weather and simple chemistry.

The Analysis tab is the workbench behind it: station observations hour by hour, the interpolation between stations with its measured error, weather-model fields, and simulations with adjustable wind and mixing. Early warning ranks every station by the trained model's forecast against a threshold and says how uncertain each forecast is. Weather and Air Quality hold the plain forecasts and the station tables. Data & Models says what data exists, what does not, and lets you train a new model; nothing trains unless you press the button.

Where pollution is likely to increase next

There are three ways the application answers this, and they are deliberately kept apart because they have different strengths.

A statistical forecast for each station. A model trained on a year of hourly observations and weather predicts PM2.5, PM10, NO2 and O3 at +1, +6 and +24 hours. On a held-out period it was closer than simply repeating the last value in eleven of twelve cases, and clearly better at six hours (PM2.5 error 15 against 18 ug/m3). It does not beat that baseline for PM10 a day ahead, and the application says so wherever that forecast appears. This is the most reliable answer to "what will this monitor read", but it needs fresh observations, and the station feed is currently days behind.

A physical picture of where the observed air goes. The transport and coupled models carry the current field with the wind and apply deposition, rain scavenging and sulphate formation. They show where polluted air is heading, which is exactly the question of regional movement. They have no emissions, so they cannot say where new pollution will appear, and on historical windows they did not beat simply holding the starting value. Read them as a picture of movement, not as a number to rely on.

An independent atmospheric model. CAMS (the Copernicus atmospheric composition service, reached through Open-Meteo) is a full chemistry model with emissions. It is updated every hour and covers the whole area, but its cells over Delhi are about 44 km across and, compared with the stations over the last two months, it was a worse predictor than "the same hour yesterday" (PM2.5 error 30 against 18.5 ug/m3; correlation 0.29). It is useful as regional context and as a check, not as a station forecast.

Who is exposed

Exposure is concentration multiplied by the number of people breathing it, weighted towards the sensitive: children, the elderly, people in hospitals and schools. The application does not yet compute this, because the two layers it needs are not loaded: a gridded population (WorldPop or GHSL are free) and a list of hospitals, schools and care homes. The database schema already has tables for both, and the Data Coverage page lists them as obtainable. Until they are loaded, any map that claimed to show "who is affected" would be inventing it, so none does.

How to read the colours and labels

Every layer carries a badge that says what kind of number it is. Observed is a measurement at a station. Spatial estimate is an interpolation between stations and is clear wherever fewer than two stations are within 12 km. Weather forecast and Atmospheric model are model output, not measurements. Model forecast is the trained model's prediction for one station. Simulated and Hypothetical are computed from assumptions, and scenarios are what-ifs rather than predictions. A Stale badge means the newest data are older than the limit; the application then describes the past and says so.

What is missing

The honest list of absent data is on the Data Coverage page and in the data-sources report. The ones that matter most for the questions above are an emissions inventory, fire detections, nitrogen oxide and station-weather observations (offered by a feed already in use but not yet stored), vertical profiles of pollutants, and the population and sensitive-site layers. Several can be added without cost; the reports say how.

Where the data comes from, and how far to trust it

> *A live audit of the three sources the application relies on (Open-Meteo with CAMS, OpenAQ, and the xKDR archive), run on 11 October 2026. Every figure below was observed in a response on that day; where a fact could not be verified it says so. The audit found a six-hour timestamp error in one source, which has been fixed.*

The short version

Source	What it gives	Newest data	Resolution	Main caveat
Open-Meteo forecast	Weather: wind, rain, temperature, boundary-layer height, radiation, winds up to 3 km	now, to 16 days ahead	cells about 8 x 10 km	Model output, never a measurement
Open-Meteo archive	The same, as historical analysis, from 1940	about 23 hours ago	9 km (IFS) / 0.25° (ERA5)	One-day lag
CAMS via Open-Meteo	Modelled PM2.5, PM10, NO2, SO2, O3, CO, NO, dust, AOD	now, to 7 days ahead	cells about 0.4° (44 km)	A model; on Delhi stations worse than "same hour yesterday"
OpenAQ	Hourly station observations, 115 monitors in NCR	median 80 hours ago; 1 of 113 within 3 hours	points	Feed lags by days; licences mostly unstated

Source	What it gives	Newest data	Resolution	Main caveat
xKDR archive	CPCB and US Embassy hourly history, 2009 onward	CPCB: 1 Sep 2025; US Embassy: 27 Mar 2026	points	Timestamp convention differs by source

The two sentences that matter most: there is no live ground truth for Delhi at the moment (the observation feed trails the present by more than three days), and the only current pollutant field is a coarse model. Everything the application shows as "now" is therefore either a replay, a model, or a simulation, and each is labelled as such.

Open-Meteo: weather

The forecast endpoint returned 552 hourly rows for a request of seven past days and sixteen forecast days, with no missing values in ten variables. It accepts up to sixteen days ahead (seventeen is refused with HTTP 400) and more than ninety days of past data. A request for Delhi was snapped to the grid point at 28.576° N, 77.187° E at 214 m elevation, and probing at 0.02° spacing shows the native cells are about 0.070° tall and 0.103° wide (8 x 10 km). Wind speed is returned in km/h unless metres per second are requested, and the application asks for metres per second throughout; conversions are recorded rather than silent.

The archive reaches back to 1940 and its newest hour was 23 hours old. Neither endpoint returns rate-limit headers, so the limit can only be seen by hitting it. Open-Meteo's published free allowance is 600 calls a minute, 5,000 an hour and 10,000 a day for non-commercial use; in practice, bursts of archive requests during the evaluation of the coupled model were refused and needed spacing and a retry. Licence: the data are CC BY 4.0, the free service is non-commercial, and commercial use needs a subscription.

The application also reads winds at 80, 120 and 180 m and on the 925, 850 and 700 hPa surfaces (about 0.8, 1.5 and 3.2 km above ground), which is what makes the vertical structure of the 3D flow real rather than decorative.

CAMS: the only current pollutant field

Open-Meteo serves CAMS global composition forecasts. A 92-day past window plus seven days ahead was returned with 1.5 % missing values; eight days ahead is refused. Units are ug/m3 (aerosol optical depth is dimensionless). Stepping 0.05° at a time across 1.1° of longitude gave only three distinct values, changing at 77.0° and 77.4°, so the effective cell is 0.4° or about 44 km and Delhi-NCR holds only about nine cells. This is why the live field on the Air Flow page is made of large blocks, and why no city-scale structure should be read into it.

How good is it at the stations? For the 60 days ending 9 October 2026 (the newest stored hour), CAMS at 60 stations' coordinates was compared with the station hourly values, at the best of a few time shifts:

Pollutant	Obs. mean (ug/m3)	CAMS bias	CAMS MAE	Same hour yesterday MAE	Hour-of-day mean MAE	Correlation
PM2.5	43	+20	30.0	18.5	19.3	0.29
PM10	116	+34	91.7	43.3	48.9	0.26
NO2	52	-19	31.5	15.3	15.9	0.23
O3	24	+71	73.1	7.9	9.2	0.44
SO2	31	-6	18.9	8.8	10.8	0.02

CAMS is a poor predictor of an individual Delhi station here: for every pollutant it is further from the observation than repeating yesterday's value, and its correlation is low. Two cautions. This compares the model's current estimate for past hours, not a forecast issued a day earlier. And the large ozone bias deserves a unit and siting check before it is read as a model error alone (stations near traffic destroy ozone that a 44 km cell cannot see). The finding that matters for design is the first one: a raw CAMS value is not a substitute for a station forecast, which is why a local model is justified.

OpenAQ: the observations

OpenAQ's version 3 API requires a key, allows 60 requests a minute (the response headers say so: limit 60, remaining counted down), and returned 115 reference-grade monitors inside the NCR box, 82 from CPCB, 30 from the CAAQM network and a handful from AirNow and the State Air programme. Each location lists its sensors with parameters and units.

The audit found three things worth knowing. Freshness is poor: the time since each location's last datum had a median of about 80 hours; one location was within three hours and 24 hours; 39 of 113 had not reported for more than a week, some for years. Licences are mostly unstated: one location listed a licence (US Public Domain); 114 listed none, so

redistribution terms cannot be assumed and the application shows OpenAQ's attribution. Units are mixed: gases arrive as ppb, ppm or ug/m3, depending on the sensor, which is why every value is converted with a documented factor (25 °C, 1 atm) and keeps its original value and unit beside it.

OpenAQ also carries NO, NOx, temperature, relative humidity, wind speed and wind direction on CPCB stations. None of these is ingested yet. That corrects an earlier statement in this project that nitrogen oxide and station weather were unavailable: they are available, from a feed already in use.

The hourly-values response has value, a period with UTC and local start and end, a coverage block (expected and observed counts and percentage), and flagInfo. Periods are labelled as the provider gives them; for CPCB stations they end at the top of the Indian hour, which is why their UTC times fall on the half hour.

xKDR: the long history, and a six-hour error

The xKDR archive holds 196.5 million hourly rows from 558 stations since January 2009, 72 of them inside the NCR box, exported on 30 September 2026. It is a history, not a feed: CPCB stations in Delhi-NCR end on 1 September 2025, and only the US Embassy monitor continues, to 27 March 2026. Its metadata says timestamps are naive Indian Standard Time.

The audit tested that claim against the one station that both xKDR and OpenAQ serve over the same months. Over 1,306 overlapping hours the two series are identical (correlation 1.000, mean absolute difference 0.0 ug/m3), but only when xKDR's times are shifted by six hours relative to how the application had been reading them. A physical check agrees: read as IST, the Embassy series has its daily minimum at 10:00, when the afternoon boundary layer should make it 16:00 to 17:00; read as UTC it falls where it should. The other xKDR stations are correct as documented: across 40 CPCB stations the median daily minimum falls at 17:00 in the xKDR period and 16:00 in the OpenAQ period.

So the US Embassy rows are in UTC and the rest in IST. The ingest now reads that source as UTC, a regression test pins the behaviour, and the affected station (OpenAQ id 8118, the one the live forecast was served from) was re-ingested and cleaned: its daily minimum moved from 11:00 to 17:00. The previous raw rows are backed up. The trained model predates this fix, so retraining is recommended (it is a button on the Model Training page; nothing was retrained here).

Quality flags and the stored record

Every stored value keeps its original value and unit, the converted value, and a list of flags: low coverage, unit converted, unit suspect, out of range, flat-lined, sensor conflict, revised, and others. Values that fail a physical-plausibility check are kept and flagged, not clipped or dropped. Carbon monoxide is excluded entirely because every CO sensor in the window is labelled ppb with a median near 1, which is physically impossible; the labels, not the air, are wrong.

A single typed record now describes every value the application handles: its kind (observation, forecast, analysis, interpolation or simulation), the exact period it describes in UTC, when it was issued and retrieved, the station or the model cell and its size, original and standard units, flags, licence and attribution. The kind is part of a record's identity, so an interpolated value can never overwrite a measured one. The database section of the engineering report has the schema.

What this means in practice

For "what is the air like right now", the CAMS model is the only current source and is labelled as a model. For "what did the monitors read", the store is accurate but days old. For "what will this monitor read", the trained model needs fresh observations the feed does not currently supply, which is why it declines to forecast for the present. Closing these gaps does not need a new model: it needs a fresher observation feed (or direct CPCB access), the NO/NOx and station-weather parameters that OpenAQ already offers, and a retraining after the timestamp repair.

Which models, and why: an assessment before any new training

> *Candidates for station forecasting, regional prediction, weather and wind, transport and exposure, compared on what they need, what they give and what they cost. Only some were run; the report says which, and does not claim results for the rest. The recommendation is a layered system built on what already works, not a new large model.*

What was asked, and the rule followed

The questions are practical: how pollutants affect air quality across the region, how weather and transport move them, where pollution is likely to rise next, and who is exposed. The rule for answering was to measure the cheap options first. Nothing large was downloaded and no dependency was installed for this assessment; the numbers below come from the data already in the project and from live calls to the public sources.

What was measured

Four baselines and one trained family were evaluated on the same stations and hours, so the comparison is fair.

Approach	PM2.5 +1 h	PM2.5 +6 h	PM2.5 +24 h	PM10 +24 h	NO2 +6 h	O3 +6 h
Persistence (repeat the last value)	8.7	17.9	18.5	41.4	18.5	14.3
Trained ensemble (RF, LightGBM, XGBoost, SVR)	8.5	15.1	17.5	42.4	12.9	7.7
Raw CAMS at the station (60-day window)	n/a	n/a	30.0	91.7	31.5	73.1
Passive transport simulation (25 windows, 12 h)	worse than persistence					
Coupled model, deposition + washout + oxidation	worse than persistence					

Mean absolute error in ug/m3 on the held-out period from late July to early October 2026, on 68 to 70 stations. The CAMS row is not tied to a horizon (it is a single hourly estimate) and sits in the column of its pollutant; the two simulation rows were scored over 25 historical windows of 12 hours at withheld stations. The ensemble is closer than persistence in eleven of twelve target and horizon cells, by a lot at six hours (PM2.5 is about 16 % better, O3 about 46 % better), and by little at one hour, where yesterday's and the last hour's values are already nearly right. It does not beat persistence for PM10 a day ahead, and the application flags that cell wherever it appears. Prediction intervals were fitted from validation residuals and cover 84 to 94 % of held-out values at a nominal 80 %, which is honest but slightly wide.

Three findings shaped everything else. First, persistence is a hard baseline in a source-poor model: air quality has strong memory, and any method that cannot add information beyond the last observation will lose to it. Second, raw CAMS is not a station forecaster here: it is biased and weakly correlated at 44 km cells. Third, the physical models (transport and coupled) are valuable for understanding movement and for visualisation, and are not forecasters, because they contain no emissions.

The candidates

Candidate	Inputs	Output and scale	Hardware	Licence	Complexity	Verdict
Persistence, hour-of-day climatology	station history	station, any horizon	none	n/a	trivial	Keep as the baseline every model must beat
Gradient-boosted trees and forests (in use)	79 features: lags, rolling stats, weather, time, neighbours	station, +1/+6/+24 h, intervals	CPU, minutes	permissive (BSD/Apache/MIT)	low	Keep and extend
Boosted trees with CAMS and weather forecasts as features	the above plus CAMS and Open-Meteo forecast values at the station	station, up to +72 h	CPU	permissive	low	Recommended next experiment

Candidate	Inputs	Output and scale	Hardware	Licence	Complexity	Verdict
Quantile gradient boosting / conformal intervals	same	calibrated intervals	CPU	permissive	low	Recommended: replaces ad hoc residual bands
Temporal fusion / N-HiTS (neural time series)	station histories, covariates	multi-horizon, quantiles	single GPU helpful	Apache/MIT	medium	Defer: little expected gain over trees on 70 stations and one year
Graph or ConvLSTM spatiotemporal nets	station graph or gridded fields	regional field, hours ahead	GPU	BSD/MIT	high	Defer: needs a gridded target; 70 stations are sparse
CAMS global composition forecast (in use as context)	none (served)	0.4°, hourly, 5 days	none	Copernicus, free with attribution	none	Use as regional background and as a feature, not alone
FourCastNet (NVlabs)	ERA5 initial state, 20 variables	global weather, 0.25°, 6-hourly, no chemistry	GPU (A100 class for speed)	code BSD-style; check weights terms	high	Reject for this use: coarser than the 9 km weather already served, no pollutants
Foundation models with chemistry heads (for example Aurora)	ERA5/CAMS states	global fields at 0.1 to 0.4°	GPU, large weights	check model card	high	Not evaluated; interesting later as a CAMS-like prior, not needed now
Passive transport (in use)	wind, boundary layer	2.7 km field, hours	CPU, seconds	own code	low	Keep for visualisation and what-ifs
Lagrangian trajectories (NOAA HYSPLIT, FLEXPART)	meteorological archives	back-trajectories to source regions	CPU	free / GPL	medium	Recommended: the right tool for "where did this smoke come from"
WRF-Chem or similar	emissions, boundary data	1 to 10 km chemistry	HPC cluster	open	very high	Out of scope without an emission inventory
Population-weighted exposure	concentration field, population raster, site list	exposure per area and group	CPU	WorldPop CC BY	low once the data are loaded	Recommended (data not loaded yet)

Items marked "not evaluated" were assessed from their published descriptions only, not run, and nothing here claims a result for them.

Why not FourCastNet, specifically

FourCastNet is a data-driven global weather emulator. It forecasts atmospheric variables such as wind at 0.25° (about 28 km) every six hours from an ERA5 starting state. For this project that is the wrong tool in three ways: Open-Meteo already serves hourly, 9 km forecasts from operational models at no cost; it has no pollutant channels, so it cannot say anything about concentration; and running it needs ERA5 initial conditions and a GPU. A justified weather use would be a regional ensemble of winds with quantified uncertainty, which is a research project, not a prerequisite. It is therefore not installed and its weights were not downloaded.

Recommended architecture

The system is five layers, each of which exists or can be built from what is already here.

1. Data layer. One typed, normalised record for every value (observation, forecast, analysis, interpolation, simulation), stored in PostgreSQL with PostGIS where available and SQLite for development. Provenance, units, periods, quality flags and licences travel with every row. This is implemented; the migration is 0002.

2. Observation and background fusion. Station observations interpolated between stations, with CAMS as the regional background and boundary condition. The interpolation's leave-one-out error is shown; CAMS is never presented as a measurement.

3. Station forecaster. Gradient-boosted models per pollutant and horizon, with CAMS and weather forecasts at the station as additional inputs, evaluated against persistence and climatology with the same time-ordered split. The one experiment worth running next is the CAMS-feature ablation: does adding CAMS reduce the 6 h and 24 h error? It is cheap and decides whether any heavier model is needed.

4. Movement and what-if. The transport and coupled models, labelled as simulations, for showing where the air goes and for scenario questions, with trajectories added for source attribution.

5. Exposure. Population-weighted concentration and a sensitive-site layer, produced from the station forecast and the interpolated field, once the population raster and site list are loaded.

How each model is evaluated

Every candidate is run on time-ordered splits with the training period strictly before the validation period, which is strictly before the test period; features are computed by the same function in training and serving, with a latency parameter, so a feature can never see the future. Scores are MAE, RMSE, R2 and bias per pollutant and horizon, stratified by season and concentration band where the sample allows, always beside persistence and climatology. A candidate is adopted per cell only where it beats persistence on held-out data, and cells that do not are flagged in the interface. Intervals are checked for coverage. Spatial claims are tested leave-one-station-out. Simulations are scored against later observations at stations withheld from their starting field.

What would change the recommendation

A fresh, gap-free observation feed would make the station forecaster usable for the present and would let the physical models be started from measurements. An emissions inventory would turn the transport model into something that can forecast concentration changes rather than only carry them. A gridded observation product (satellite aerosol) would make a spatiotemporal network worth training. Until then the cheapest, most defensible improvements are the data repairs, the missing parameters, retraining on the corrected series, and the CAMS-feature ablation.

Engineering report: defects found, fixes, setup and what remains

> *What was wrong in the data pipeline and the web application, in priority order, with the root cause and the evidence for each fix; then how to set the system up, migrate the database and re-run every check. Nothing is marked as passing that was not run.*

Inventory

The backend is FastAPI on Python 3.13 (the project's pins also work on 3.14), with SQLite for the historical store (15.5 GB, 68 stations), Chroma for the knowledge base and Alembic scaffolding for PostgreSQL/PostGIS. The frontend is Next.js 16.4.0 with Turbopack, React 19.3, MapLibre 6.13, deck.gl 9.4, three 0.185 with React Three Fiber 9.8, and ECharts 6. The build uses Next's Cache Components and partial prefetching; every page is a client component, and the production build prerenders all routes statically. PostgreSQL and PostGIS are not configured on this machine (the status page reports both unreachable, as it should), so the new schema is delivered as a migration and a SQL file rather than applied.

Defects, in priority order

#	Severity	Where	Root cause	Fix	Evidence
1	High (data correctness)	api/app/history/ingest_xkdr.py	xKDR's US-Embassy rows were read as Indian time but are UTC; the series sat 5.5 to 6 hours off the same station's OpenAQ data, and was merged into station 8118, the one the live forecast was served from	Sources listed in UTC_NAIVE_SOURCES are read as UTC; the station was re-ingested, re-cleaned and its features rebuilt by app/history/repair_xkdr_utc.py, with a backup	Series identical at +6 h (r 1.000, MAE 0.0, 1,306 h); daily minimum 11:00 -> 17:00; test test_us_embassy_timestamps_are_read_as_utc_not_ist
2	High (data correctness)	api/app/services/data_availability.py	The coverage catalogue asserted that NO, NOx and station weather were unavailable; the audit found OpenAQ serves them	Catalogue corrected and a distinct "obtainable" status added	docs/phase1/source-audit.json; tests/test_data_availability.py
3	Medium (documentation of a limit)	api/app/services/cams_fields.py and its callers	CAMS resolution was stated as about 11 km; measured it is 0.4° (44 km), so the live field is a few blocks	Text corrected everywhere it appeared	Step probe in the audit: 3 distinct values over 1.1° of longitude
4	Medium (frontend)	web/src/components/twin/scene3d/Scene3D.tsx	drei's <Html> mounts a separate React root per label and unmounts it synchronously during React 19's commit; under StrictMode this raised removeChild (121 times) and createRoot (88 times) errors on every mount of the 3D view	Labels are now canvas-texture sprites (TextSprite in ground.tsx); no DOM roots exist inside the WebGL tree	Dev log counts before; 0 errors captured when opening 3D after
5	Medium (frontend)	web/src/components/charts/EChart.tsx	The chart was initialised inside hidden tabs and collapsed panels while their width was 0 ("Can't get DOM width or height", four times)	The chart is created when its container first has a size	Warnings absent after change
6	Medium (accuracy)	api/app/services/weather_fields.py	The weather lattice (8 x 9) under-sampled the model's native cells (0.07° x 0.10°)	Lattice raised to 14 x 11, matching the native grid	Probe at 0.02° spacing
7	Medium (modelling)	api/app/twin/coupled.py	The one-sided dilution rule and, later, the ventilation rule lowered skill in a 25-window hindcast	Both are opt-in; the evidence is shown in the interface	docs/coupled-evaluation.md

#	Severity	Where	Root cause	Fix	Evidence
8	Low (security)	api/app/api/training.py	State-changing training endpoints could be reached by a cross-site form	Require header X-AEROCAST-Action: training	test_state_changing_training_calls_need_the_action_header
9	Low (performance)	api/app/api/twin.py	Early warning recomputed 40 station forecasts (about a minute) on every request	Per-station forecasts cached by pollutant, origin, stations and active model	test_early_warning_reuses_station_forecasts...
10	Low (interface)	web/src/components/map/useBaseMap.ts	One dark basemap, no imagery or terrain; a flat 3D ground	Voyager, Dark, Satellite and Relief; real terrain and imagery in 3D	docs/terrain-and-imagery.md

Not defects, but worth knowing

The browser console in development mode still shows THREE.Clock: This module has been deprecated, which comes from inside @react-three/fiber 9.8 and is harmless; it should disappear with an update of that library. Next.js prints "Route ... is rendering with server caches disabled" in development, an informational message of the Cache Components feature. The OpenAQ feed's lag and the CPCB archive's end in September 2025 are properties of the sources, not of the code.

Setup

Backend (from api):

```
python -m venv .venv .venv\Scripts\python.exe -m pip install -r requirements-ml.txt copy .env.example .env .venv\Scripts\python.exe -m uvicorn app.main:app --port 8000
```

api/.env keys: AQ_API_KEY (OpenAQ), INDIA_AQI_API_KEY (xKDR), OLLAMA_MODEL, ML_LOCAL_MODEL_DIR=./data/models, CORS_ORIGINS, and DATABASE_URL only when PostgreSQL is used. Frontend (from web): npm install, then npm run dev; web/.env.local takes NEXT_PUBLIC_CARTO_API_KEY (public by design; restrict it to your domains).

Database: schema and migration

The normalised schema is in docs/phase1/schema.postgis.sql and in Alembic revision 0002_environmental_schema (after 0001, which enables PostGIS). It has: env_source (provider, licence, attribution, native resolution, cadence); env_station and env_cell (point and polygon geometries with GiST indexes); env_record (one row per value, with kind constrained to observation, forecast, analysis, interpolation or simulation, a CHECK that forecasts carry an issue time and observations do not, and a CHECK that a record without a value is not usable); and the two exposure layers exposure_cell and sensitive_site, which are empty until a population raster and a facility list are loaded.

To apply it on a PostgreSQL server with PostGIS available: set DATABASE_URL in api/.env, then run .venv\Scripts\python.exe -m alembic upgrade head from api. The role needs permission to create the extension or PostGIS must already be installed. Rollback is alembic downgrade 0001. The development store keeps the same typed record in SQLite (app/envdata/schema.py, EnvStore), so code written against it runs unchanged against the PostgreSQL table.

Re-running every check

```
# backend: unit and integration tests (external services are mocked) cd api && .venv\Scripts\python.exe -m pytest -q # source audit and cross-checks against the live providers (a few dozen requests) .venv\Scripts\python.exe -m app.envdata.audit .venv\Scripts\python.exe -m app.envdata.crosscheck_xkdr .venv\Scripts\python.exe -m app.envdata.source_eval # the hindcast of the coupled model (about 25 minutes, rate-limited by the weather API) .venv\Scripts\python.exe -m app.twin.coupled_eval --from 2026-08-04T00:00 --to 2026-10-04T00:00 --step-hours 60 --hours 12 # frontend cd web && npx tsc --noEmit && npx eslint src && npx vitest run && npm run build
```

Results of the run described in this report: backend 794 passed; frontend 163 passed in 19 files; tsc and eslint clean; the production build compiled and prerendered all 15 routes; npm audit --omit=dev found 0 vulnerabilities in production dependencies (development tooling was not audited). One backend test failed once in a heavily loaded full run and

passed in every later run, alone and in the full suite; it is recorded as intermittent rather than explained.

Remaining limitations

The observation feed is days old, so the trained model refuses to forecast for the present and every "now" map is a replay or a model. The trained model was fitted before the timestamp repair and should be retrained (a user action). Raw CAMS is coarse (44 km) and, over Delhi stations, a worse predictor than yesterday's value. The coupled model has no emissions; its couplings did not reliably improve on plain transport and nothing beat persistence. The 3D concentration colouring inside the mixed layer assumes a uniform concentration with height, because there are no vertical measurements. Station licences in OpenAQ are mostly unstated, so redistribution terms are unknown. NO, NOx and station weather are available but not yet ingested. Population and sensitive-site layers are not loaded, so exposure is not computed. PostgreSQL and PostGIS are not configured here, so the migration is untested against a live server; its DDL is covered only by string-level tests. The model candidates marked as not evaluated in the model report were not run.